



1. Description

1.1. Project

Project Name	CatFeeder
Board Name	NUCLEO-G474RE
Generated with:	STM32CubeMX 6.17.0
Date	07/03/2026

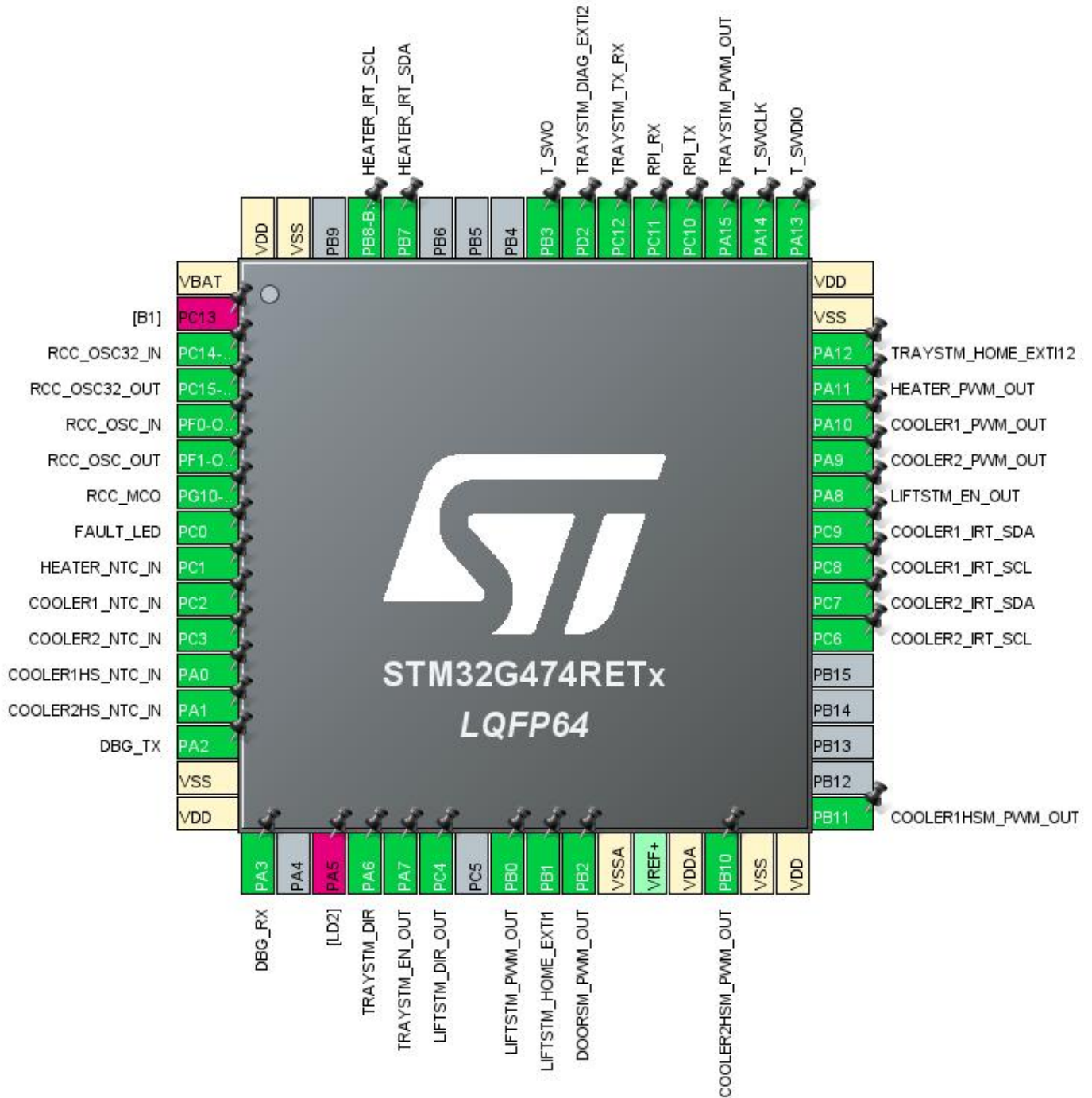
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x4
MCU name	STM32G474RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

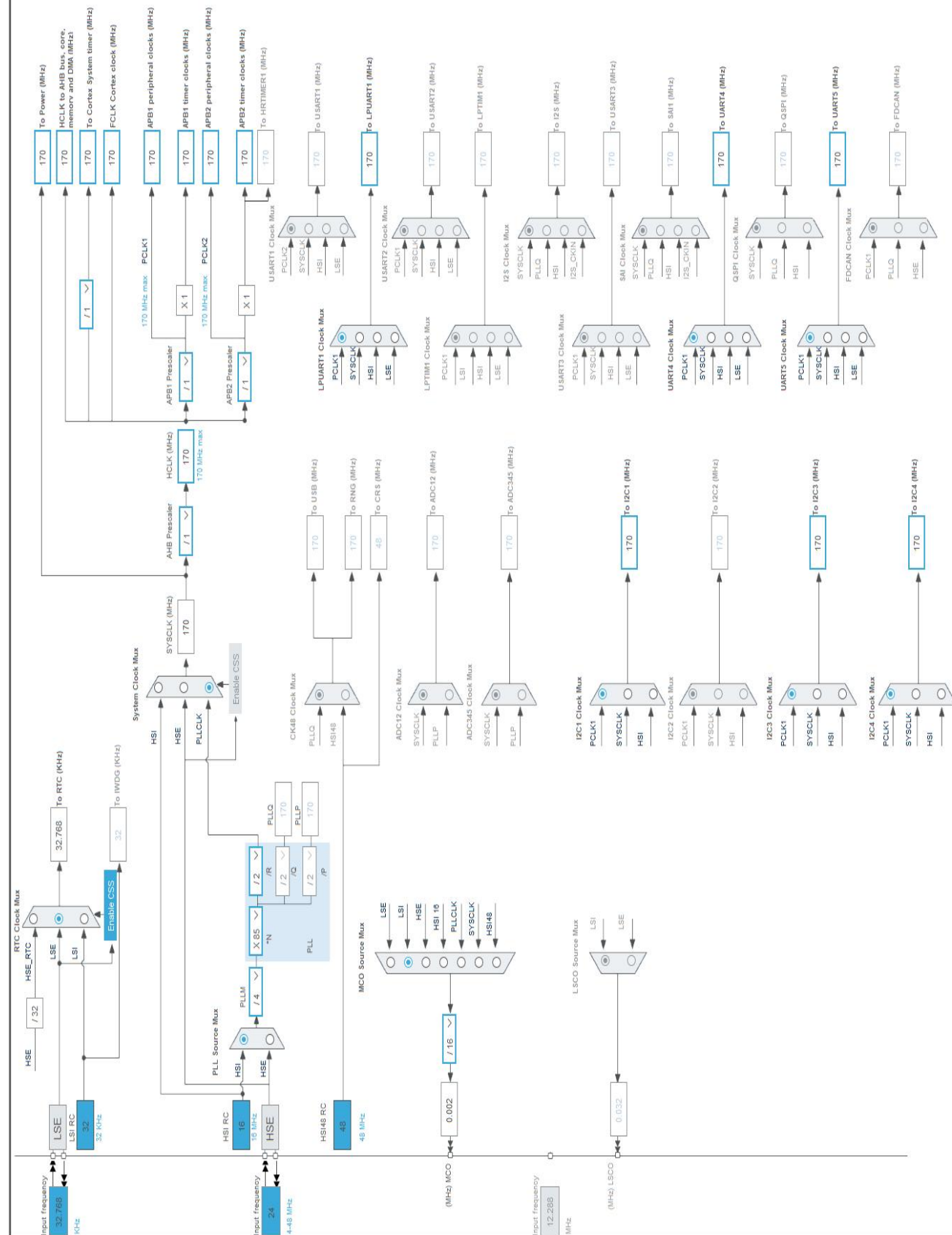
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13	I/O		
3	PC14-OSC32_IN	I/O	RCC_OSC32_IN	RCC_OSC32_IN
4	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	RCC_OSC32_OUT
5	PF0-OSC_IN	I/O	RCC_OSC_IN	RCC_OSC_IN
6	PF1-OSC_OUT	I/O	RCC_OSC_OUT	RCC_OSC_OUT
7	PG10-NRST	I/O	RCC_MCO	
8	PC0 *	I/O	GPIO_Output	FAULT_LED
9	PC1	I/O	ADC1_IN7	HEATER_NTC_IN
10	PC2	I/O	ADC1_IN8	COOLER1_NTC_IN
11	PC3	I/O	ADC1_IN9	COOLER2_NTC_IN
12	PA0	I/O	ADC1_IN1	COOLER1HS_NTC_IN
13	PA1	I/O	ADC1_IN2	COOLER2HS_NTC_IN
14	PA2	I/O	LPUART1_TX	DBG_TX
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	LPUART1_RX	DBG_RX
19	PA5	I/O		
20	PA6 *	I/O	GPIO_Output	TRAYSTM_DIR
21	PA7 *	I/O	GPIO_Output	TRAYSTM_EN_OUT
22	PC4 *	I/O	GPIO_Output	LIFTSTM_DIR_OUT
24	PB0	I/O	TIM3_CH3	LIFTSTM_PWM_OUT
25	PB1	I/O	GPIO_EXTI1	LIFTSTM_HOME_EXTI1
26	PB2	I/O	TIM5_CH1	DOORM_PWM_OUT
27	VSSA	Power		
29	VDDA	Power		
30	PB10	I/O	TIM2_CH3	COOLER2HSM_PWM_OUT
31	VSS	Power		
32	VDD	Power		
33	PB11	I/O	TIM2_CH4	COOLER1HSM_PWM_OUT
38	PC6	I/O	I2C4_SCL	COOLER2_IRT_SCL
39	PC7	I/O	I2C4_SDA	COOLER2_IRT_SDA
40	PC8	I/O	I2C3_SCL	COOLER1_IRT_SCL
41	PC9	I/O	I2C3_SDA	COOLER1_IRT_SDA
42	PA8 *	I/O	GPIO_Output	LIFTSTM_EN_OUT
43	PA9	I/O	TIM1_CH2	COOLER2_PWM_OUT

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
44	PA10	I/O	TIM1_CH3	COOLER1_PWM_OUT
45	PA11	I/O	TIM1_CH4	HEATER_PWM_OUT
46	PA12	I/O	GPIO_EXTI12	TRAYSTM_HOME_EXTI12
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	T_SWDIO
50	PA14	I/O	SYS_JTCK-SWCLK	T_SWCLK
51	PA15	I/O	TIM8_CH1	TRAYSTM_PWM_OUT
52	PC10	I/O	UART4_TX	RPI_TX
53	PC11	I/O	UART4_RX	RPI_RX
54	PC12	I/O	UART5_TX	TRAYSTM_TX_RX
55	PD2	I/O	GPIO_EXTI2	TRAYSTM_DIAG_EXTI2
56	PB3	I/O	SYS_JTDO-SWO	T_SWO
60	PB7	I/O	I2C1_SDA	HEATER_IRT_SDA
61	PB8-BOOT0	I/O	I2C1_SCL	HEATER_IRT_SCL
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32G474RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x4
MCU	STM32G474RETx
Datasheet	DS12288 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

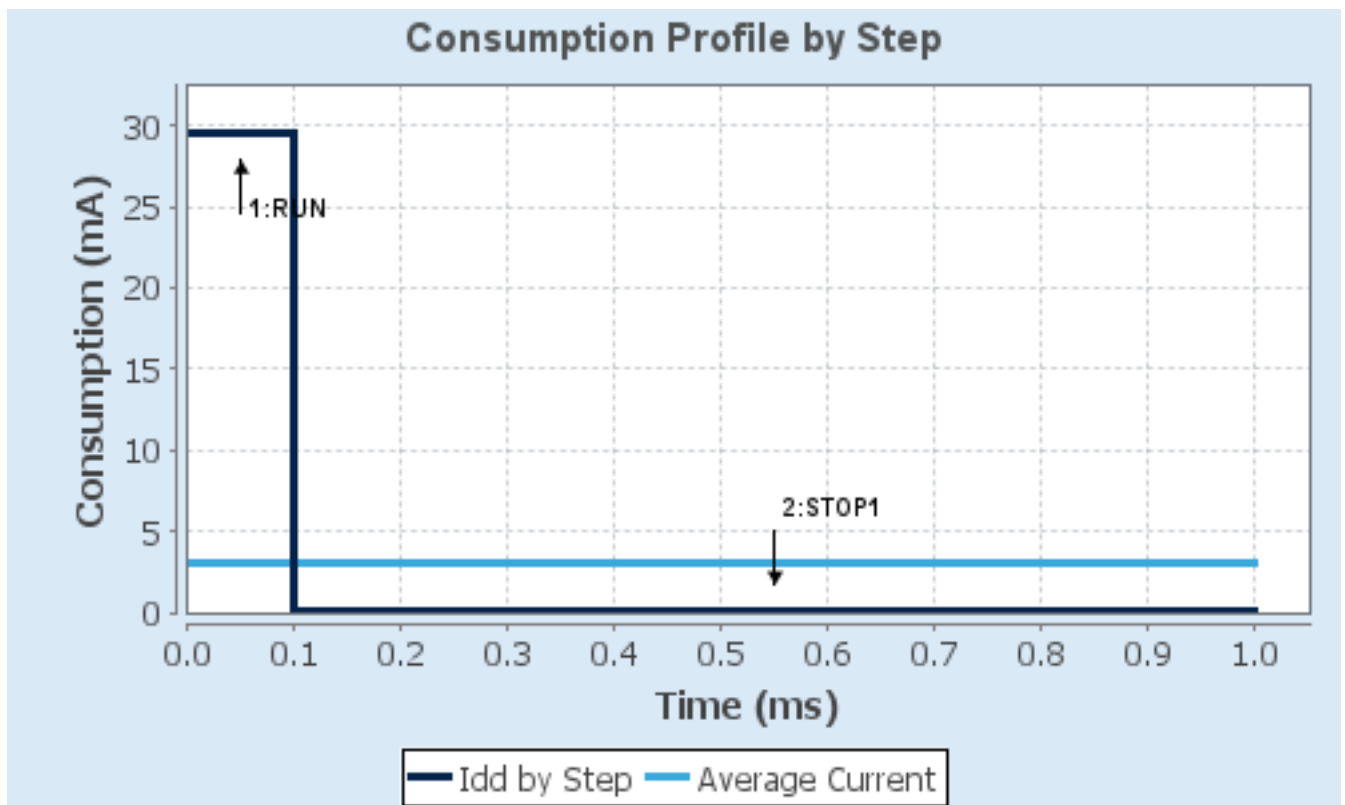
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	CatFeeder
Project Folder	C:\Users\Bowen\Documents\GitHub\CatFeeder_Firmware\CatFeeder
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.2
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	Yes

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_DMA_Init	DMA
3	SystemClock_Config	RCC
4	MX_LPUART1_UART_Init	LPUART1
5	MX_RTC_Init	RTC
6	MX_CRC_Init	CRC
7	MX_ADC1_Init	ADC1
8	MX_I2C1_Init	I2C1
9	MX_UART4_Init	UART4
10	MX_TIM1_Init	TIM1
11	MX_TIM8_Init	TIM8

Rank	Function Name	Peripheral Instance Name
12	MX_UART5_Init	UART5
13	MX_I2C3_Init	I2C3
14	MX_I2C4_Init	I2C4
15	MX_TIM2_Init	TIM2
16	MX_TIM3_Init	TIM3
17	MX_TIM5_Init	TIM5

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN7: IN7 Single-ended

IN8: IN8 Single-ended

IN9: IN9 Single-ended

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Enabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion **7 ***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 1

Sampling Time 2.5 Cycles

Offset Number No offset

Rank **2 ***

Channel **Channel 2 ***

Sampling Time 2.5 Cycles

Offset Number No offset

<u>Rank</u>	3 *
Channel	Channel 7 *
Sampling Time	2.5 Cycles
Offset Number	No offset
<u>Rank</u>	4 *
Channel	Channel 8 *
Sampling Time	2.5 Cycles
Offset Number	No offset
<u>Rank</u>	5 *
Channel	Channel 9 *
Sampling Time	2.5 Cycles
Offset Number	No offset
<u>Rank</u>	6 *
Channel	Channel Temperature Sensor *
Sampling Time	2.5 Cycles
Offset Number	No offset
<u>Rank</u>	7 *
Channel	Channel Vrefint *
Sampling Time	2.5 Cycles
Offset Number	No offset
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. CRC

mode: Activated

3.2.1. Parameter Settings:

Basic Parameters:

Default Polynomial State	Enable
Default Init Value State	Enable

Advanced Parameters:

Input Data Inversion Mode	None
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Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.3. I2C1

I2C: I2C

3.3.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40B285C2 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.4. I2C3

I2C: I2C

3.4.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40B285C2 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.5. I2C4

I2C: I2C

3.5.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40B285C2 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.6. LPUART1

Mode: Asynchronous

3.6.1. Parameter Settings:

Basic Parameters:

Baud Rate	460800 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
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Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Disable *
DMA on RX Error	Enable
MSB First	Disable

3.7. NUCLEO-G474RE

mode: Human Machine Interface

3.7.1. Human Machine Interface:

Led:

USER LED GREEN (LD2)	true *
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Button:

USER B1	Mode EXTI *
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VCOM:

Virtual Com Port	false
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Demonstration code:

Generate demonstration code	Enabled
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3.8. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

mode: Master Clock Output

3.8.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled

Prefetch Buffer	Enabled *
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)
RCC Parameters:	
HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost
Peripherals Clock Configuration:	
Generate the peripherals clock configuration	TRUE

3.9. RTC

mode: Activate Clock Source

3.9.1. Parameter Settings:

General:	
Hour Format	Hourformat 24
Asynchronous Predivider value	127
Synchronous Predivider value	255

3.10. SYS

Debug: Trace Asynchronous Sw

Timebase Source: TIM7

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.11. TIM1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4

3.11.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	0
Counter Mode	Up

Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.12. TIM2

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4

3.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.13. TIM3

Channel3: PWM Generation CH3

3.13.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.14. TIM5

Channel1: PWM Generation CH1

3.14.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.15. TIM8

Channel1: PWM Generation CH1

3.15.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.16. UART4

Mode: Asynchronous

3.16.1. Parameter Settings:

Basic Parameters:

Baud Rate	1000000 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.17. UART5

Mode: Single Wire (Half-Duplex)

3.17.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.18. FREERTOS

Interface: CMSIS_V2

3.18.1. Config parameters:

API:

FreeRTOS API	CMSIS v2
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Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	2.1.3

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Enabled *

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled

USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Enabled *

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	16000 *
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Enabled *
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Option2 *

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled

USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.18.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.18.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
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Project settings (see parameter description first):

Use FW pack heap file	Enabled
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*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC1	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	HEATER_NTC_IN
	PC2	ADC1_IN8	Analog mode	No pull-up and no pull-down	n/a	COOLER1_NTC_IN
	PC3	ADC1_IN9	Analog mode	No pull-up and no pull-down	n/a	COOLER2_NTC_IN
	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	COOLER1HS_NTC_IN
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	COOLER2HS_NTC_IN
I2C1	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	HEATER_IRT_SDA
	PB8-BOOT0	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	HEATER_IRT_SCL
I2C3	PC8	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	COOLER1_IRT_SCL
	PC9	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	COOLER1_IRT_SDA
I2C4	PC6	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	COOLER2_IRT_SCL
	PC7	I2C4_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	COOLER2_IRT_SDA
LPUART1	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DBG_TX
	PA3	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	DBG_RX
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	RCC_OSC32_IN
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	RCC_OSC32_OUT
	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	RCC_OSC_IN
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	RCC_OSC_OUT
	PG10-NRST	RCC_MCO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	T_SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	T_SWCLK
	PB3	SYS_JTDO-SWO	n/a	n/a	n/a	T_SWO
TIM1	PA9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	COOLER2_PWM_OUT
	PA10	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	COOLER1_PWM_OUT
	PA11	TIM1_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	HEATER_PWM_OUT
TIM2	PB10	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	COOLER2HSM_PWM_OUT
						T

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB11	TIM2_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	COOLER1HSM_PWM_OUT
TIM3	PB0	TIM3_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LIFTSTM_PWM_OUT
TIM5	PB2	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	DOORSM_PWM_OUT
TIM8	PA15	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	TRAYSTM_PWM_OUT
UART4	PC10	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RPI_TX
	PC11	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	RPI_RX
UART5	PC12	UART5_TX	Alternate Function Open Drain	No pull-up and no pull-down	Low	TRAYSTM_TX_RX
GPIO	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FAULT_LED
	PA6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRAYSTM_DIR
	PA7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRAYSTM_EN_OUT
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LIFTSTM_DIR_OUT
	PB1	GPIO_EXTI1	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	LIFTSTM_HOME_EXTI1
	PA8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LIFTSTM_EN_OUT
	PA12	GPIO_EXTI12	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	TRAYSTM_HOME_EXTI12
	PD2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	TRAYSTM_DIAG_EXTI2

4.2. DMA configuration

DMA request	Stream	Direction	Priority
LPUART1_RX	DMA1_Channel1	Peripheral To Memory	Low
LPUART1_TX	DMA1_Channel2	Memory To Peripheral	Low
UART5_TX	DMA1_Channel3	Memory To Peripheral	Medium *
UART5_RX	DMA1_Channel4	Peripheral To Memory	Medium *
UART4_RX	DMA1_Channel5	Peripheral To Memory	High *
UART4_TX	DMA1_Channel6	Memory To Peripheral	High *
ADC1	DMA1_Channel7	Peripheral To Memory	Medium *

LPUART1_RX: DMA1_Channel1 DMA request Settings:

Mode: **Circular ***
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

LPUART1_TX: DMA1_Channel2 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

UART5_TX: DMA1_Channel3 DMA request Settings:

Mode: Normal
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

UART5_RX: DMA1_Channel4 DMA request Settings:

Mode: **Circular ***

Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

UART4_RX: DMA1_Channel5 DMA request Settings:

Mode: **Circular ***
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

UART4_TX: DMA1_Channel6 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

ADC1: DMA1_Channel7 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Half Word
Memory Data Width: Half Word

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
EXTI line1 interrupt	true	5	0
EXTI line2 interrupt	true	5	0
DMA1 channel1 global interrupt	true	5	0
DMA1 channel2 global interrupt	true	5	0
DMA1 channel3 global interrupt	true	5	0
DMA1 channel4 global interrupt	true	5	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel6 global interrupt	true	5	0
DMA1 channel7 global interrupt	true	5	0
EXTI line[15:10] interrupts	true	5	0
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	true	15	0
LPUART1 global interrupt	true	5	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
I2C1 event interrupt / I2C1 wake-up interrupt through EXTI line 23	unused		
I2C1 error interrupt	unused		
TIM8 break interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM8 update interrupt		unused	
TIM8 trigger and commutation interrupts		unused	
TIM8 capture compare interrupt		unused	
TIM5 global interrupt		unused	
UART4 global interrupt / UART4 wake-up interrupt through EXTI line 34		unused	
UART5 global interrupt / UART5 wake-up interrupt through EXTI line 35		unused	
FPU global interrupt		unused	
I2C4 event interrupt / I2C4 wake-up interrupt through EXTI line 42		unused	
I2C4 error interrupt		unused	
I2C3 event interrupt / I2C3 wake-up interrupt through EXTI line 27		unused	
I2C3 error interrupt		unused	

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	false	false
Hard fault interrupt	false	false	false
Memory management fault	false	false	false
Prefetch fault, memory access fault	false	false	false
Undefined instruction or illegal state	false	false	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
EXTI line1 interrupt	false	true	true
EXTI line2 interrupt	false	true	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
DMA1 channel7 global interrupt	false	true	true
EXTI line[15:10] interrupts	false	true	true
TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts	false	true	true
LPUART1 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware								
FREERTOS								
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Utilities	Bsp
DMA	ADC1	RTC	I2C1			CRC		NUCLEO-G474RE
GPIO		TIM1	I2C3					
NVIC		TIM2	I2C4					
RCC		TIM3	LPUART1					
SYS		TIM5	UART4					
		TIM8	UART5					

6. Docs & Resources

Type

Link